Inventory Management System

Technical Portfolio & Architecture Specification

DOMAIN: Enterprise Logistics & Supply Chain
TECH STACK: Python, MySQL, Pandas, Matplotlib

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1. System Overview & Problem Statement

Modern retail and logistics networks require deterministic, transactional accuracy for high-volume stock movements. Conventional spreadsheet-based tracking fails to provide transaction isolation, leading to race conditions, inventory drift, and inaccurate demand forecasting.

This Inventory Management System addresses these operational bottlenecks by providing transactional guarantees, automated threshold auditing, and integrated analytical pipelines.

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2. Technical System Architecture

- **CLI / Application Layer:** `main.py`
- **Business Logic & Operations:** `operations.py`
- **Data Access Layer:** ACID Transactions via MySQL Engine (`database.py`)
- **Analytics Engine:** Pandas Pipeline & Matplotlib Visualizations (`analytics.py`)

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3. Key Architecture Principles

- **ACID-Compliant Transactions:** Enforces strict isolation and atomicity across stock additions, order fulfillments, and supplier returns.
- **Analytical Intelligence:** Integrated data pipelines using Pandas and Matplotlib for demand analytics and velocity mapping.
- **Auditability & Traceability:** Immutable logging of stock updates and dynamic reorder threshold alerts.

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4. Database Schema Design (DDL)

```
CREATE DATABASE IF NOT EXISTS inventory_db;

USE inventory_db;

CREATE TABLE IF NOT EXISTS suppliers (
    supplier_id INT AUTO_INCREMENT PRIMARY KEY,
    name VARCHAR(100) NOT NULL,
    contact_email VARCHAR(100) UNIQUE,
    phone VARCHAR(20)
);

CREATE TABLE IF NOT EXISTS products (
    product_id INT AUTO_INCREMENT PRIMARY KEY,
    sku VARCHAR(50) UNIQUE NOT NULL,
    name VARCHAR(100) NOT NULL,
    category VARCHAR(50),
    unit_price DECIMAL(10, 2) NOT NULL,
    stock_quantity INT DEFAULT 0,
    reorder_threshold INT DEFAULT 10,
    supplier_id INT,
    FOREIGN KEY (supplier_id) REFERENCES suppliers(supplier_id) ON DELETE SET NULL
);

CREATE TABLE IF NOT EXISTS inventory_transactions (
```

```
    transaction_id INT AUTO_INCREMENT PRIMARY KEY,
    product_id INT NOT NULL,
    transaction_type ENUM('INBOUND', 'OUTBOUND', 'ADJUSTMENT') NOT NULL,
    quantity INT NOT NULL,
    timestamp TIMESTAMP DEFAULT CURRENT_TIMESTAMP,
    FOREIGN KEY (product_id) REFERENCES products(product_id)
);
```

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5. Engineering Practices & Execution Strategy

- **Separation of Concerns:** Modular architecture isolates database connectivity, business rules, and analytical reporting into dedicated sub-modules.
- **Secure Environment Configuration:** Decoupled sensitive database credentials using standardized environment variable parameters (`.env.example`).
- **Data Integrity & Constraints:** Foreign key constraints and atomic transaction handling prevent partial writes during concurrent order processing.
- **Automated Reorder Alerts:** Threshold detection logic dynamically triggers system alerts when SKU quantities drop below safe operational bounds.